Rafi Dahoui

(438) 836-3772 | dahoui.rafi27@gmail.com | rafi-dahoui.vercel.app
linkedin.com/in/rafi-dahoui-711b31148 | github.com/doctor277

Education

Université de Montréal **Expected April 2027**
Bachelor of Science (B.Sc.) in Computer Science *Montréal, QC*

Projects

LaunchGuard | Deployment monitoring platform | Source *Java, Spring Boot, Kafka, React, PostgreSQL, Docker*

- Built an **event-driven monitoring platform** that opens an incident when a service fails and records which deployment was live at the time
- Designed a **Kafka pipeline** where a worker runs HTTP health checks and a Spring Boot API deduplicates results, stores history in PostgreSQL, and resolves incidents on recovery
- Secured the dashboard with **OIDC and role-based access**; tested with **Testcontainers** and Playwright in CI

GhostWatch | Real-time Windows monitoring tool | Source *Python, FastAPI, SQLite, WebSockets, Playwright*

- Built a **local-first Windows dashboard** with FastAPI and WebSockets that tracks system load, processes, network connections, and Event Log data in real time
- Implemented **threshold and baseline anomaly detection** that shows why each alert fired, plus SQLite telemetry history and privacy-redacted reports
- Open source (MIT) with a synthetic Demo Mode; CI runs **600+ Pytest tests** and Playwright browser tests

Pothole Montréal | Civic mobile app | Source *React Native, TypeScript, Supabase, PostGIS, Next.js*

- Built a **bilingual (English/French) React Native app** to report potholes with a photo, GPS position, and map pin
- Wrote **PostGIS geospatial queries** that flag existing reports within 25 metres, so users can attach to a known pothole instead of filing a duplicate
- Built a **Supabase backend** with row-level security and private photo storage, plus a Next.js admin dashboard with server-side authorization for moderation

Investment Research Lab | Backtesting platform *Python, Pytest, SQLite*

- Built a **Python backtesting platform** that tests trading strategies against SPY with historical data
- Implemented **walk-forward validation**, transaction costs, and train/validation/test splits to limit look-ahead bias
- Added an **AI research-agent pipeline** that generates and critiques strategy ideas, covered by **1,271 automated tests**

Technical Skills

Programming: Java, Python, TypeScript, JavaScript, SQL, C, PHP
Frameworks & APIs: Spring Boot, FastAPI, React, React Native, Next.js, Node.js, REST APIs
Data & Infrastructure: PostgreSQL, PostGIS, MongoDB, Supabase, SQLite, Kafka, Docker, AWS
Testing & Tools: JUnit, Pytest, Playwright, Testcontainers, Git, GitHub Actions, Maven
Languages: English, French, Arabic

Work Experience

3 Amigos **Oct 2022 – Present**
Waiter *Montréal, QC*

- Handle **10+ tables** during high-volume shifts, keeping orders accurate and service on time
- Coordinate with kitchen and bar teams to keep orders moving and resolve customer issues as they come up
- **Train new waiters** and close the restaurant at the end of the night

Grillades Torino **Oct 2021 – Oct 2022**
Manager *Montréal, QC*

- Oversaw **daily operations** and coordinated staff to keep service moving through peak hours
- Built the **staff schedule**, trained new employees, and managed inventory and supplier orders
- Handled cash and **end-of-day reconciliation**